

Lecture 6

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Convention. The ground set is $E = \{e_1, \dots, e_n\}$. For $e \notin S$, write

$$f(e \mid S) = f(S + e) - f(S),$$

and let $[a]_+ = \max\{a, 0\}$. Unless stated otherwise, $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$ is submodular but need not be monotone or normalized. For vectors, $x \vee y$ denotes the coordinatewise maximum.

1 Double Greedy

We consider unconstrained nonmonotone submodular maximization

$$\max\{f(S) : S \subseteq E\}. \quad (1)$$

Random-half sampling ignores the values observed during the algorithm. Double greedy instead processes the elements one at a time and compares the gain from accepting an element with the gain from rejecting it. Fix an ordering $E = \{e_1, \dots, e_n\}$ and an optimal set OPT.

At the beginning of iteration i , the algorithm maintains $A_{i-1} \subseteq B_{i-1}$, where A_{i-1} consists of elements already accepted and B_{i-1} contains all elements not yet rejected. In particular, $e_i \in B_{i-1} \setminus A_{i-1}$. Define the two raw gains

$$a_i = f(e_i \mid A_{i-1}), \quad b_i = f(B_{i-1} \setminus \{e_i\}) - f(B_{i-1}) = -f(e_i \mid B_{i-1} \setminus \{e_i\}). \quad (2)$$

Since $A_{i-1} \subseteq B_{i-1} \setminus \{e_i\}$, diminishing returns gives

$$a_i + b_i \geq 0. \quad (3)$$

1.1 Deterministic double greedy

Algorithm 1 Deterministic double greedy

Input: A value oracle for f and an ordering $E = (e_1, \dots, e_n)$.

Initialize: $A_0 \leftarrow \emptyset$ and $B_0 \leftarrow E$.

for $i = 1, \dots, n$ **do**

$a_i \leftarrow f(A_{i-1} + e_i) - f(A_{i-1})$.

$b_i \leftarrow f(B_{i-1} - e_i) - f(B_{i-1})$.

if $a_i \geq b_i$ **then**

$A_i \leftarrow A_{i-1} + e_i$ and $B_i \leftarrow B_{i-1}$.

else

$A_i \leftarrow A_{i-1}$ and $B_i \leftarrow B_{i-1} - e_i$.

end if

end for

return $S \leftarrow A_n$; at termination, $A_n = B_n$.

Theorem 1.1 (Deterministic double greedy). *The returned set satisfies*

$$3f(S) \geq f(\text{OPT}) + f(\emptyset) + f(E).$$

In particular, deterministic double greedy is a 1/3-approximation [3].

Proof. Couple the algorithm to the hybrid set

$$O_i = (\text{OPT} \cup A_i) \cap B_i. \tag{4}$$

Then $O_0 = \text{OPT}$, $O_n = S$, and $A_i \subseteq O_i \subseteq B_i$.

Suppose first that the algorithm accepts e_i . If $e_i \notin O_{i-1}$, then $O_i = O_{i-1} \cup \{e_i\}$, and submodularity gives

$$\begin{aligned} f(O_{i-1}) - f(O_i) &= -f(e_i \mid O_{i-1}) \\ &\leq -f(e_i \mid B_{i-1} \setminus \{e_i\}) = b_i \leq a_i. \end{aligned}$$

If $e_i \in O_{i-1}$, the hybrid set does not change. In either case, its loss is at most $a_i = f(A_i) - f(A_{i-1})$. Notice that $a_i \geq 0$ in this branch: otherwise $a_i \geq b_i$ would contradict (3).

Now suppose the algorithm rejects e_i . The hybrid set changes only when $e_i \in O_{i-1}$. In that case,

$$\begin{aligned} f(O_{i-1}) - f(O_i) &= f(e_i \mid O_{i-1} \setminus \{e_i\}) \\ &\leq f(e_i \mid A_{i-1}) = a_i \leq b_i. \end{aligned}$$

Here $b_i \geq 0$, since $b_i > a_i$ and $a_i + b_i \geq 0$. We have therefore proved, in either branch, that

$$f(O_{i-1}) - f(O_i) \leq f(A_i) - f(A_{i-1}) + f(B_i) - f(B_{i-1}). \tag{5}$$

Summing (5) and using $O_0 = \text{OPT}$ and $O_n = A_n = B_n = S$ gives

$$f(\text{OPT}) - f(S) \leq f(S) - f(\emptyset) + f(S) - f(E).$$

Rearranging proves the theorem. □

1.2 Randomized double greedy

Put

$$\alpha_i = [a_i]_+ = \max\{a_i, 0\}, \quad \beta_i = [b_i]_+ = \max\{b_i, 0\}.$$

It is important to distinguish the raw gains a_i, b_i from their nonnegative parts α_i, β_i .

Algorithm 2 Randomized double greedy

Input: A value oracle for f and an ordering $E = (e_1, \dots, e_n)$.

Initialize: $A_0 \leftarrow \emptyset$ and $B_0 \leftarrow E$.

for $i = 1, \dots, n$ **do**

$a_i \leftarrow f(A_{i-1} + e_i) - f(A_{i-1})$ and $b_i \leftarrow f(B_{i-1} - e_i) - f(B_{i-1})$.

$\alpha_i \leftarrow [a_i]_+$ and $\beta_i \leftarrow [b_i]_+$.

if $\alpha_i + \beta_i = 0$ **then**

$p_i \leftarrow 1$ and $q_i \leftarrow 0$.

else

$p_i \leftarrow \alpha_i / (\alpha_i + \beta_i)$ and $q_i \leftarrow \beta_i / (\alpha_i + \beta_i)$.

end if

 Draw $U_i \sim \text{Unif}[0, 1]$, independently of earlier draws.

if $U_i \leq p_i$ **then**

$A_i \leftarrow A_{i-1} + e_i$ and $B_i \leftarrow B_{i-1}$.

else

$A_i \leftarrow A_{i-1}$ and $B_i \leftarrow B_{i-1} - e_i$.

end if

end for

return $S \leftarrow A_n$; at termination, $A_n = B_n$.

For the analysis, define O_i by (4). Let $\mathbb{E}_i$ denote expectation over the choice at iteration i , conditioned on all earlier choices.

Lemma 1.2 (One-step comparison). *At every iteration,*

$$\begin{aligned} \mathbb{E}_i[f(O_{i-1}) - f(O_i)] &\leq \frac{1}{2} \mathbb{E}_i[f(A_i) - f(A_{i-1}) \\ &\quad + f(B_i) - f(B_{i-1})]. \end{aligned}$$

Proof. First suppose $\alpha_i + \beta_i > 0$. The expected increase in the two endpoints is

$$p_i a_i + q_i b_i = \frac{\alpha_i^2 + \beta_i^2}{\alpha_i + \beta_i}. \quad (6)$$

Indeed, a branch whose raw gain is negative has probability zero.

If $e_i \in O_{i-1}$, only rejection changes the hybrid set. Its loss in that branch is at most $a_i \leq \alpha_i$, so its expected loss is at most $q_i \alpha_i$. If $e_i \notin O_{i-1}$, only acceptance changes the hybrid set. Its loss is at most $b_i \leq \beta_i$, so its expected loss is at most $p_i \beta_i$. Thus, in either case,

$$\mathbb{E}_i[f(O_{i-1}) - f(O_i)] \leq \frac{\alpha_i \beta_i}{\alpha_i + \beta_i} \leq \frac{1}{2} \frac{\alpha_i^2 + \beta_i^2}{\alpha_i + \beta_i},$$

where the last inequality is $2\alpha_i \beta_i \leq \alpha_i^2 + \beta_i^2$.

If $\alpha_i + \beta_i = 0$, then $a_i, b_i \leq 0$. Together with (3), this implies $a_i = b_i = 0$. The endpoint gain is zero and the hybrid loss is nonpositive, so the conclusion still holds. $\square$

Theorem 1.3 (Randomized double greedy). *The returned set satisfies*

$$\mathbb{E}[f(S)] \geq \frac{1}{2} f(\text{OPT}) + \frac{1}{4} (f(\emptyset) + f(E)). \quad (7)$$

Consequently, randomized double greedy is a 1/2-approximation [3].

Proof. Take expectations in Lemma 1.2 and sum over all iterations. The two sides telescope to

$$f(\text{OPT}) - \mathbb{E}[f(S)] \leq \frac{1}{2}(2\mathbb{E}[f(S)] - f(\emptyset) - f(E)).$$

Rearranging gives (7). The final approximation ratio uses the nonnegativity of $f(\emptyset)$ and $f(E)$. $\square$

Both double-greedy algorithms make only a linear number of value-oracle queries. The $1/2$ factor cannot be improved by a polynomial-query value-oracle algorithm in general [2].

2 Continuous Greedy

We now return to the multilinear extension F defined earlier. If f is monotone, the first-derivative identity implies $\nabla F(x) \geq 0$, so F is coordinatewise monotone. Concavity along nonnegative directions also gives the tangent inequality

$$F(y) - F(x) \leq \langle \nabla F(x), y - x \rangle \quad \text{whenever } 0 \leq x \leq y \leq \mathbf{1}. \quad (8)$$

Let $P \subseteq [0, 1]^E$ be a nonempty, convex, down-closed polytope over which linear functions can be optimized. We compare against an integral optimal solution:

$$\mathbf{1}_{\text{OPT}} \in P, \quad f(\text{OPT}) = \max\{f(S) : \mathbf{1}_S \in P\}.$$

Algorithm 3 Ideal continuous greedy

Input: The multilinear extension F with exact gradient access;

a down-closed polytope P with exact linear optimization; and $T \in [0, 1]$.

Initialize: $x(0) \leftarrow 0$.

for continuous time $t \in [0, T]$ **do**

Choose, with a fixed measurable tie-breaking rule, $v(t) \in \arg \max_{v \in P} \langle v, \nabla F(x(t)) \rangle$.

Evolve x according to $\dot{x}(t) = v(t)$ almost everywhere.

end for

return $x(T)$.

Algorithm 3 is an ideal continuous-time process, rather than a finite oracle algorithm; discretization is discussed in Section 6.

The trajectory is feasible. Indeed, for $t > 0$,

$$\frac{x(t)}{t} = \frac{1}{t} \int_0^t v(s) ds \in P$$

by convexity. Since $t \leq 1$ and P is down-closed, $x(t) \in P$. More explicitly, $x(t) = t(x(t)/t) \leq x(t)/t$ coordinatewise.

Theorem 2.1 (Continuous greedy). *If f is monotone, the ideal process with $T = 1$ satisfies*

$$F(x(1)) \geq \left(1 - \frac{1}{e}\right) f(\text{OPT}).$$

Proof. Let $x^* = \mathbf{1}_{\text{OPT}}$ and define

$$d(t) = (x^* - x(t))_+ = x(t) \vee x^* - x(t).$$

Since $0 \leq d(t) \leq x^*$ and P is down-closed, $d(t) \in P$. Optimality of $v(t)$ and (8) give

$$\begin{aligned} \frac{d}{dt}F(x(t)) &= \langle \nabla F(x(t)), v(t) \rangle \\ &\geq \langle \nabla F(x(t)), d(t) \rangle \\ &\geq F(x(t) \vee x^*) - F(x(t)) \\ &\geq f(\text{OPT}) - F(x(t)), \end{aligned}$$

where the last step uses monotonicity. Multiplying by e^t and integrating from 0 to t yields

$$F(x(t)) \geq (1 - e^{-t})f(\text{OPT}) + e^{-t}F(0) \geq (1 - e^{-t})f(\text{OPT}).$$

The result follows by setting $t = 1$ [4]. □

2.1 A nonmonotone stopping-time bound

For a nonmonotone objective, the last inequality in the preceding proof is unavailable. We replace it by the following estimate, used in analyses of measured continuous greedy [5].

Lemma 2.2 (Union lower bound). *Suppose f is nonnegative and submodular. If $x \in [0, 1]^E$ and $\|x\|_\infty \leq p \leq 1$, then every $S \subseteq E$ satisfies*

$$F(x \vee \mathbf{1}_S) \geq (1 - p)f(S). \tag{9}$$

Proof. The case $p = 0$ is immediate. Assume $p > 0$, set $u = (x \vee \mathbf{1}_S) - \mathbf{1}_S$, and define

$$g(\lambda) = F(\mathbf{1}_S + \lambda u), \quad 0 \leq \lambda \leq 1/p.$$

We have $0 \leq u \leq p(\mathbf{1} - \mathbf{1}_S)$, so the entire displayed segment stays in $[0, 1]^E$. The vector u is nonnegative, so g is concave. Since $1 = (1 - p) \cdot 0 + p \cdot (1/p)$,

$$F(x \vee \mathbf{1}_S) = g(1) \geq (1 - p)g(0) + pg(1/p) \geq (1 - p)f(S).$$

The final inequality uses $F \geq 0$. □

Theorem 2.3 (Early-stopped continuous greedy). *For a nonnegative, possibly nonmonotone function f , the ideal continuous- greedy process stopped at $T = \ln 2$ satisfies*

$$F(x(T)) \geq (1 - \ln 2)f(\text{OPT}).$$

Proof. Since $v(t) \in [0, 1]^E$, each coordinate satisfies $x_e(t) \leq t$. The first three lines in the proof of the monotone guarantee remain valid. Lemma 2.2, applied with $p = t$, therefore gives

$$\frac{d}{dt}F(x(t)) \geq (1 - t)f(\text{OPT}) - F(x(t)).$$

Multiplying by e^t and integrating gives

$$F(x(T)) \geq (2 - T - 2e^{-T})f(\text{OPT}) + e^{-T}F(0).$$

The coefficient $2 - T - 2e^{-T}$ is maximized on $[0, 1]$ at $T = \ln 2$, where it equals $1 - \ln 2$. □

3 Measured Continuous Greedy

We use the capped-direction formulation appearing in the handwritten notes. At a point x , the direction must belong to P and must fit inside the remaining coordinatewise capacity $\mathbf{1} - x$.

Algorithm 4 Ideal measured continuous greedy

Input: The multilinear extension F with exact gradient access;

a down-closed polytope P with exact optimization over $\{v \in P : v \leq \mathbf{1} - x\}$; and $T \in [0, 1]$.

Initialize: $x(0) \leftarrow 0$.

for continuous time $t \in [0, T]$ **do**

Choose, with a fixed measurable tie-breaking rule,

$v(t) \in \arg \max \{\langle v, \nabla F(x(t)) \rangle : v \in P, v \leq \mathbf{1} - x(t)\}$.

Evolve x according to $\dot{x}(t) = v(t)$ almost everywhere.

end for

return $x(T)$.

The feasible region is nonempty at every time because it contains 0.

As before, $x(t)/t$ is an average of points in P , so $x(t) \in P$ for $t \leq 1$. The cap in Algorithm 4 also gives

$$\dot{x}_e(t) \leq 1 - x_e(t) \quad \implies \quad x_e(t) \leq 1 - e^{-t}. \quad (10)$$

Theorem 3.1 (Measured continuous greedy). *At time $T \leq 1$, the ideal capped-direction process satisfies*

$$\begin{aligned} F(x(T)) &\geq (1 - e^{-T})f(\text{OPT}) && \text{if } f \text{ is monotone,} \\ F(x(T)) &\geq Te^{-T}f(\text{OPT}) && \text{if } f \text{ is nonnegative and possibly nonmonotone.} \end{aligned}$$

Consequently, at $T = 1$ the guarantees are $1 - 1/e$ and $1/e$, respectively.

Proof. Let $x^* = \mathbf{1}_{\text{OPT}}$ and set

$$d(t) = (x^* - x(t))_+ = x(t) \vee x^* - x(t).$$

We have $d(t) \leq x^* \in P$, so $d(t) \in P$ by down-closedness. Moreover, $d(t) \leq \mathbf{1} - x(t)$. Thus $d(t)$ is a feasible comparator for the direction choice in Algorithm 4, and

$$\begin{aligned} \frac{d}{dt}F(x(t)) &= \langle \nabla F(x(t)), v(t) \rangle \\ &\geq \langle \nabla F(x(t)), d(t) \rangle \\ &\geq F(x(t) \vee x^*) - F(x(t)). \end{aligned} \quad (11)$$

If f is monotone, the right-hand side is at least $f(\text{OPT}) - F(x(t))$, and the continuous-greedy analysis applies. If f is nonmonotone, (10) and Lemma 2.2 imply

$$F(x(t) \vee x^*) \geq e^{-t}f(\text{OPT}).$$

It follows from (11) that

$$\frac{d}{dt}(e^t F(x(t))) \geq f(\text{OPT}).$$

After integration,

$$F(x(T)) \geq e^{-T} F(0) + T e^{-T} f(\text{OPT}) \geq T e^{-T} f(\text{OPT}).$$

This proves the claims. In particular, $1/e \approx 0.3679$ [5]. □

Remark (Oracle requirement). The capped-direction presentation above requires linear optimization over the time-dependent polytope

$$\{v \in P : v \leq \mathbf{1} - x(t)\}.$$

The original measured continuous-greedy algorithm of Feldman, Naor, and Schwartz scales each coordinate of a direction from P by its remaining capacity; that formulation needs only a linear-optimization oracle for P and yields the same ideal differential bounds [5].

4 Discretization and Rounding

The exact constants above describe ideal continuous-time trajectories. A finite implementation uses the relevant linear-optimization oracle, replaces the differential equation by sufficiently small Euler steps, and estimates each coordinate of $\nabla F(x)$ by random sampling. For the capped-direction formulation, the relevant oracle is the stronger one identified in the preceding remark. For every fixed $\varepsilon > 0$, the standard discretized algorithms run in time polynomial in n and $1/\varepsilon$ under these oracles and give, in expectation, $1 - 1/e - \varepsilon$ in the monotone case and $1/e - \varepsilon$ for nonmonotone measured continuous greedy [4, 5].

The output x is fractional. Under a matroid constraint, pipage rounding converts x to an independent set without decreasing the multilinear value in expectation [6, 4]. The relevant geometric fact follows from the mixed partials computed earlier: for distinct i, j , the function

$$t \mapsto F(x + t(\mathbf{1}_i - \mathbf{1}_j))$$

is convex on every interval for which its argument stays in $[0, 1]^E$. For an arbitrary down-closed polytope, the required rounding procedure is a separate part of the algorithmic problem.

Summary. For unconstrained nonnegative submodular maximization, deterministic double greedy achieves $1/3$ and randomized double greedy achieves the optimal value-oracle factor $1/2$. For a monotone objective over a down-closed polytope, continuous greedy obtains the ideal fractional factor $1 - 1/e$. Ordinary continuous greedy gives $1 - \ln 2$ for a nonmonotone objective when stopped at time $\ln 2$, while measured continuous greedy improves the ideal factor to $1/e$. Discretization, sampling, and rounding turn the continuous trajectories into algorithms for discrete feasible sets.

References

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